

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester-III Examination, 2022-23

Botany

Paper: BOB-301 — Plant Ecology and Physiology

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following questions in not more than 50 words each: $[1 \times 10 = 10]$

- (a) What is the full form of RuBisCO?
- (b) What is apical dominance?
- (c) Name a pump found in the plasma membrane.
- (d) Define autogenic succession.
- (e) What is tropopause?
- (f) Name the precursor of auxins.
- (g) What are hydathodes?
- (h) What is the function of cholesterol in plasma membranes?
- (i) What is residual soil?
- (j) What are aquaporins?

SECTION A

Question 2. With the help of suitable diagrams, discuss the different adaptations of Xerophytes. $[15]$

Question 3. Write short notes on any two of the following: $[7.5 \times 2 = 15]$

- (a) Quantitative characters of a community
- (b) Population growth-curves
- (c) Ecological pyramids

Question 4. Discuss in detail any two of the following: $[7.5 \times 2 = 15]$

- (a) Survivorship curves
- (b) Soil profile

- (c) Effect of temperature on the distribution of plants

Question 5. Describe the different developmental stages of a hydrosere. [15]

SECTION B

Question 6. With the help of suitable diagram, describe the fluid mosaic model of the structure of cell membrane. [15]

Question 7. Write short notes on any two of the following: $[7.5 \times 2 = 15]$

- (a) Components of water potential
- (b) Physiological role of gibberellins
- (c) Mechanism of phytochrome action

Question 8. Discuss any two of the following in brief: $[7.5 \times 2 = 15]$

- (a) Secondary active transport
- (b) Glycolysis
- (c) Structure of nitrogenase

Question 9. What is phosphorylation? With the help of a suitable diagram, describe the chemiosmotic model of phosphorylation in plants. [15]

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